LI // LOGAIN INDUSTRIES

FDN

FREIGHT DISTRIBUTION NETWORK

for X4: Foundations



X4: FOUNDATIONS 9.00 // v1.00 BETA

User Guide

FDN

LI Freight Distribution Network

Logain Abler · Logain Industries

Supply resources. Return products.
Collect overstock. Connect your stations.

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Unofficial fan content. Requires a legally owned copy of X4: Foundations.
Beta: back up saves. Defaults may change during beta.

LI // FDN / START HERE

Welcome to FDN

Make player stations into **Distribution Centres (DCs)**. FDN uses their stock to supply production stations, return surplus products and collect overstock. Transfers have fees; the read-only Overview helps you monitor the network.

This is a **new X4 mod**: similar in idea to X3 FDN, but not the same mod and **not a port**. FDN does not sell directly to NPC stations; use normal in-game sell offers on your DCs.

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> BEFORE YOU BEGIN
>
> v1.00 is a beta. Back up your saves before adding or removing FDN. Defaults may change during beta.

Guide based on README/CHANGELOG revision 6b1d9ea and checked defaults, with updated pre-release packaging instructions. Workshop and forum links remain pending publication.

LI // FDN / SETUP

Requirements & installation

What you need

- **X4: Foundations 9.00.**
- **SirNukes Mod Support APIs** — required. Workshop id `2042901274`; extension folder `sn_mod_support_apis`. FDN uses its Simple Menu API and Lua loader.

FDN's extension id and folder are `fdn_x4`. Enable both extensions in the Extension Manager.

Manual zip · GitHub / GOG

1. Download `fdn_x4_v1.00.zip` from **FDN 1.00 BETA** (tag `v1.00`, GitHub pre-release): github.com/mattryan72/FDN_X4/releases
2. Extract the zip into the **game's own extensions folder**:
   `X4 Foundations\extensions\`
   This creates the top-level `fdn_x4` folder.
3. Install SirNukes Mod Support APIs into:
   `X4 Foundations\extensions\sn_mod_support_apis`
4. Enable both in the Extension Manager.

> IMPORTANT: GAME FOLDER, NOT DOCUMENTS
>
> Do **not** install FDN only in `Documents\Egosoft\X4\<id>\extensions`. The game reports Mod Support APIs as missing there. Use the game-install paths above. Workshop subscribers are unaffected.

Steam Workshop · when published

(Workshop URL pending publication)

Once the FDN item is published, subscribe to both FDN and SirNukes Mod Support APIs, then enable both. You must subscribe to the dependency yourself.

Protected UI Mode

You do **not** need to turn it off. FDN uses Mission Director scripts and one small UI Lua file that only reads, sets and clears station storage-level overrides using vanilla functions; it does not set buy/sell limits, prices or trade rules.

README.md and LICENSE ship in the extension root. This PDF is intended for GitHub distribution, not the extension package. The release download becomes available when a release is published.

LI // FDN / FIRST RUN

Quick start & station controls

1. Install FDN and its required dependency; enable both.
2. FDN automatically registers every player-owned station, including new builds.
3. Right-click a suitable player station and **buy an FDN Charter**. It becomes a Distribution Centre.
4. Review the station toggles below. Open the **FDN Overview** to check stock, fees and Needs attention.
5. Use the **SirNukes Mod Support APIs options menu** to adjust FDN settings.



Station right-click controls. Resource supply and send products to DC are the two per-station toggles; both are on by default.

Two controls per station

- **FDN: resource supply ON/OFF** controls incoming FDN resource supply.
- **FDN: send products to DC ON/OFF** controls product returns and resource overstock collection. It is hidden on DCs.

The equivalent globals are **Supply station resources** and **Send products to DC**. Precedence is **per-station setting → global setting → default**; a station override can therefore take priority over a global change.

> NO ELIGIBLE DC IN RANGE
>
> The station receives no new FDN supply and runs normally; FDN hands its stock levels back. Previously queued deliveries may still finish.

LI // FDN / DC MANAGEMENT

Distribution Centres & charters

A **Distribution Centre** is a player station with a purchased FDN Charter. It provides the stock and pays the fees for FDN's transfers. Right-click the station to buy a charter or revoke it.



Charter action on a player station. The price depends on qualifying docks and standing; this screenshot is an example, not a fixed charter price.

Eligibility and price

The station needs an operational dock area or pier from FDN's whitelist: selected **Argon, Boron and Terran dock areas**, and **Argon, Paranid, Teladi, Split, Boron and Terran piers**. Not every dock qualifies. Ship-build bays, equipment docks, Xenon and landmark parts do not qualify.

The base fee is **75% of the average price of the most expensive qualifying dock**, clamped to **1,500,000–4,000,000 Cr**, then the standing discount is applied.

You need standing **10+** with the sector owner. In player-owned, ownerless, Xenon or Kha'ak sectors, FDN uses your best standing with an eligible economy faction.

Standing discounts: 10% at standing 10; 25% at 20; 40% at 30.

> REVOKING A CHARTER
>
> Revoking ends DC status with **no refund**. For complete removal of FDN, follow the recommended uninstall procedure later in this guide.

LI // FDN / NETWORK BEHAVIOUR

Home DCs, networks & ware flows

Home DC versus network range

The nearest eligible DC by **gate jumps** is a station's **home DC** and is tried first. A station that is itself a DC is its own home DC.

Other DCs must be within the station's supply range — **5 jumps** by default — and part of the home DC's network. FDN tries them in order of distance from the home DC.

A network is a **chain of DCs each linked within 10 jumps** by default. The whole chain may span farther than 10 jumps. This is separate from the station's 5-jump supply limit. Turn **Limit supply range** off for no distance limit; the maximum-range slider runs from 1 to 30.

> NO PRIMARY DC
>
> There is no configurable primary DC. No network-size or DC-count cap is implemented in these paths. Supply from another DC goes directly to the receiving station, not via the home DC.

1 · Resource supply

DC stock covers **2 production cycles** by default. If stock is short, **NPC purchase fallback** is on by default. Its price cap is the home DC's buy price, or the receiving station's buy price if the home DC has no usable buy price.

2 · Product returns

Surplus product above the cycle-based keep goes to a DC; a per-ware override is available. **Product cycles to keep = 0** means no product keep. **Smallest product batch = 100** is a minimum transfer batch, **not 100 units retained**.

3 · Overstock collection

Surplus resource collection follows **Send products to DC**. For pure resources, the default trigger is **10% above the station target** (`cargo.target`).

Dual-use/self-made wares that the station also consumes have a separate cushion: collect only **above 2× need**, leave **1.5× need**, and do not collect within **10 minutes** of an FDN delivery. This avoids repeatedly sending the same stock back and forth.

Illegal wares

For returns and collection, FDN may use another legal DC in range if the ware is illegal at the first destination. If no legal destination is available, it stays at the station.

LI // FDN / ECONOMICS

Fees, delays & stock levels

Ordinary supply fee

Fees are value- and distance-based, with adjustable **flat multipliers** for volume and ware type. With shipped defaults:

```
Fee = transfer value × 0.02
  × (1 + 0.2 × gate jumps)
  × 1.0 volume factor × 1.0 ware type factor
```

Do not read “volume factor” as an extra multiplication by cargo volume. These are flat factors you can tune; the default is 1.0 for each.

Transfer	Multiplier / payer
Resource supply	Ordinary supply fee; paid by the DC
Product return	0.5× the ordinary supply fee; paid by the DC
Overstock collection	1× the ordinary supply fee; paid by the DC
Supply from another DC	Normally 0.1× the ordinary supply fee, paid by the home DC. If it cannot pay, the source DC may pay the full supply fee.

Worked example · illustration

A transfer valued at **100,000 Cr** over **3 jumps**, with both flat factors at 1.0, gives an ordinary supply fee of **3,200 Cr**: 100,000 × 0.02 × 1.6. The corresponding return multiplier gives 1,600 Cr; collection gives 3,200 Cr. This illustrates the formula, not the settings in the screenshot.

Optional delay

Transfers are instant by default. **Delay transfers by distance** is off; when enabled, the default delay is **8 seconds per jump**. Let queued transfers finish before removing FDN.

What happens to stock levels?

FDN saves original levels and sets storage-level overrides for supplied resources. It restores those originals when a ware is no longer needed. It never sets product levels and leaves self-made wares to the game. Each DC reserves **10% of each ware** by default.

> SELLING SURPLUS
>
> Use normal in-game sell offers on a DC. FDN never sells directly to NPC stations.

LI // FDN / MONITORING

Reading the FDN Overview

The **FDN Overview is read-only**. Use it to understand the current network; change settings through the Mod Support APIs options menu and station actions.



FDN Overview: network status and fees, Needs attention, per-DC stock versus target and paged recent activity. This is an example game, not a benchmark or default configuration.

A useful checking order

1. **DCs online and totals:** check that your chartered stations are represented.
2. **Needs attention:** inspect problems before changing fees or range.
3. **Per-DC stock versus target:** identify shortfalls and reserves.
4. **Network fees:** compare 1-hour, 6-hour and 24-hour totals, split into supply, return and collection.
5. **Recent activity:** page through recent moves to see what FDN has been doing.

> TWO DIFFERENT HISTORY LIMITS
>
> The DC ledger defaults to **200 entries per DC** (settings range 50–1000). Recent activity is a **separate 50-entry list**. These are not an overall bound on FDN's save data.

If stock is not moving

Check station toggles, the nearest eligible DC, station supply range, DC stock and ability to pay. Then check Overview warnings and `fdn.log`; see Troubleshooting for a concise checklist.

LI // FDN / CONFIGURATION

Settings & saved overrides

Open FDN's settings in the **SirNukes Mod Support APIs options menu**. The following two pages list shipped first-start values, not values from an existing save.



Settings page from a tuned game — values shown are not the shipped defaults. For example, this save shows a 1.5% transfer fee, 10-jump maximum supply range and 5% overstock trigger; defaults are 2.0%, 5 jumps and 10%.

Why your values can be different

Changed global settings are stored in the save and kept across updates. A new default does not overwrite an existing saved override. The menu shows your current values.

Reset all FDN settings

This clears saved **global** overrides and returns them to shipped defaults. It does **not** clear per-station toggles. Check those separately if a station behaves differently from the global setting.

BETA DEFAULTS

Defaults may change during beta. All values can be changed in the FDN settings menu. The tables use on-screen labels; the separate dual-use collection rules are not on the settings page.

"Maximum supply range" (5 jumps) and "DC network range" (10 jumps) are separate controls. To remove the station distance limit, switch Limit supply range off rather than trying to set its slider to zero.

LI // FDN / SETTINGS REFERENCE / 1

Defaults: supply & operations

Shipped first-start values for **v1.00 BETA**. Defaults may change during beta. All values can be changed in the FDN settings menu.

On-screen setting	Default	Meaning / range
NPC purchase fallback	On	NPC buys when DC stock is short
Supply check interval	60 s	Supply-check scheduling
Stations per supply pass	3	Stations processed per pass
Minimum gap between checks	20 s	Minimum check spacing
Production cycles to cover	2	Resource coverage
Supply station resources	On	Global; per-station override takes priority
Send products to DC	On	Global; also gates overstock collection
Overstock trigger	10%	0–100%, step 5; pure resources above target
Product cycles to keep	0	No product keep
Smallest product batch	100	0–5000, step 50; minimum transfer, not retained stock
DC network range	10 jumps	1–30; DC-to-DC link distance
DC reserve	10%	Each ware held at the DC
DC ledger size	200	Entries per DC; range 50–1000
Limit supply range	On	Switch off for no distance limit
Maximum supply range	5 jumps	1–30; station-to-DC reach
Delay transfers by distance	Off	Instant transfers by default
Delay per jump	8 s	Used when distance delay is on

Rules outside this settings page

Dual-use collect trigger: 200% of need. **Dual-use keep:** 150% of need. Collection is also held for 10 minutes after an FDN delivery.

Per-station defaults: resource supply ON and send products to DC ON. The products toggle is hidden on DCs. Per-station settings override global settings.

> DO NOT CONFUSE THESE VALUES
>
> A minimum product batch of **100** does not retain 100 units. A **10-jump DC link** does not make the station's default supply range 10 jumps.

LI // FDN / SETTINGS REFERENCE / 2

Defaults: fees, charters & logs

Continued shipped settings reference. “Return fee” and “DC to DC fee” are **multipliers**, not percentages of transfer value.

On-screen setting	Default	Meaning / range
Transfer fee	2.0%	0–10%, step 0.5
Fee increase per jump	0.2	Distance coefficient; up to 1
Fee volume factor	1.0	Flat multiplier; 0–3
Fee ware type factor	1.0	Flat multiplier; 0–3
Return fee	0.5	Supply-fee multiplier; 0–1
DC to DC fee	0.1	Supply-fee multiplier; 0–1; payer fallback applies
Charter fee	75%	Qualifying dock value; before clamp/ discount
Charter fee minimum	1,500,000 Cr	Clamp before standing discount
Charter fee maximum	4,000,000 Cr	Clamp before standing discount
Charter minimum standing	10	Sector owner; special-sector rule applies
Charter discount 1 at standing	10	Discount 1 = 10%
Charter discount 2 at standing	20	Discount 2 = 25%
Charter discount 3 at standing	30	Discount 3 = 40%
Logbook detail	2	Range 0–3
Notify when a DC cannot pay	On	Payment warning
Repeat skip entries after	600 s	Deduped skip-warning repeat interval
Write FDN debug log	On	Writes fdn.log
Debug log detail	3	Levels 1–4; use 4 for bug reports

Overstock collection fee: 1× the ordinary supply fee. The home DC normally pays 0.1× for pulls from another DC; the source may pay the full supply fee if the home DC cannot pay.

Charter discounts: the menu also exposes the corresponding discount percentages (10%, 25%, 40%). Revoking a charter never refunds the fee.

The charter value is based on the most expensive qualifying dock’s average price, not the value of the whole station. See the charter section for the whitelist summary and standing exceptions.

LI // FDN / DIAGNOSTICS

Warnings, logs & bug reports

FDN provides logbook notices, low-DC-stock warnings and `fdn.log`. Debug logging is on by default at detail **3**; the logbook detail default is **2**.



> Example FDN logbook notices. Use the debug log for detailed diagnosis and restoration messages; logbook detail and debug log detail are separate settings.

Why warnings do not repeat constantly

Skip warnings are deduplicated by **station + ware + kind + reason**. A changed fingerprint or a successful move can allow another warning. The default repeat-skip interval is 600 seconds.

Preparing a bug report

1. Set **Debug log detail** (log level) to **4** in FDN settings.
2. Reproduce the problem, then collect:
   `Documents\Egosoft\X4\<id>\fdn.log`
3. Include your **X4 version, DLC and FDN v1.00 BETA**, what you did, what you expected and what went wrong.

Use the Egosoft forum thread once published: **(forum thread URL pending publication)**. The GitHub project link is on the final page.

> RESTORATION CHECKS
>
> During removal, use `fdn.log` to confirm **RESTORED** (or already-original) messages and **SOLO** for managed stations losing service. Investigate restoration/helper failures before disabling FDN.

Save safety & recommended uninstall

Back up first

Adding FDN to an existing save creates its data on load. Keep a backup before adding FDN, making major configuration changes or removing it.

The DC ledger defaults to **200 entries per DC** (UI range 50–1000); recent activity keeps 50 entries. **There is no overall save-size bound.** Not every saved table or pending-transfer list has a fixed cap. Setting `LedgerMax` to zero or below outside the UI bypasses ledger trimming; keep ledger changes within the settings UI.

> WHY NOT JUST DISABLE IT?
>
> FDN sets real station storage levels. Those levels stay in the save if the mod is simply removed. Let the enabled mod restore the originals first.

Recommended removal procedure

This is the **recommended procedure**; it is not a guarantee of a tested uninstall outcome for every save.

1. **Back up your save.**
2. **Leave FDN enabled and revoke every charter.** Use **FDN: revoke charter (no refund)**. Optionally switch resource supply off. Remember that per-station switches override global settings.
3. **Keep both FDN and Mod Support APIs enabled until saved levels are restored.** Check `fdn.log` for **RESTORED** (or already-original) messages and **SOLO** for each managed station losing service. Resolve restoration/helper failures and inspect the original manual/automatic storage levels. Do not rely on a fixed waiting time.
4. **Let queued deliveries finish** if distance delay was enabled.
5. **Save to a new slot, disable FDN, then load.** Keep the original backup.

Revoking one DC is not full removal

Revoking only ends that station's DC status and gives no refund. Other charters and FDN's station management may remain active. For complete removal, work through all stations and restoration checks above.

> IF RESTORATION FAILS
>
> Do not disable the helper dependency to force removal. Keep the extensions enabled, inspect the log and gather a level-4 bug report so the failure can be diagnosed.

LI // FDN / REFERENCE

Troubleshooting & FAQ

“Mod Support APIs is missing”, but I installed it

For a manual install, check that both extensions are in the **game’s own** `X4 Foundations\extensions` folder, not Documents. Check the folders `fdn_x4` and `sn_mod_support_apis`, and enable both in the Extension Manager. Workshop subscribers are unaffected by the Documents-folder issue.

My station is not receiving resources

Check the station’s resource-supply toggle first, then the global switch. Is there an eligible DC within the supply range (default 5 jumps)? Does it have stock and money for fees? Check Overview’s Needs attention and `fdn.log`. A DC network link is not a substitute for the station’s range check.

Why are products or overstock staying at a station?

Check **Send products to DC**, product keep and the minimum batch. Pure-resource collection uses the target plus the overstock trigger. Dual-use wares need more than 2× need and a 10-minute gap after an FDN delivery. If the ware is illegal at a DC, another legal in-range DC may be tried; otherwise it stays put.

Why does the settings screenshot differ from my menu?

It shows a tuned save. Saved overrides persist across updates. Use the defaults tables for shipped values; Reset all FDN settings resets global overrides, not per-station toggles.

Why can’t I buy a charter?

Check the qualifying operational dock/pier whitelist and standing requirement (10+ by default). Not every dock is eligible. Ship-build bays and equipment docks do not qualify. The charter section explains special-sector standing rules.

Can I choose a primary DC or sell directly to NPCs?

No primary DC is configurable: the nearest eligible DC is home. There is no explicit DC-count/network-size cap in these paths. Sell DC surplus through normal in-game sell offers; FDN does not sell directly to NPC stations.

Do I need to disable Protected UI Mode?

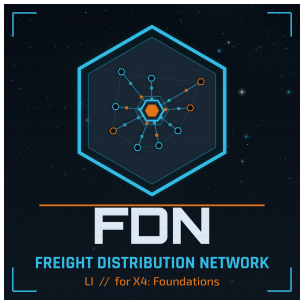
No. FDN’s small Lua helper only handles storage-level overrides with vanilla functions. It does not set trade rules or buy/sell limits.

> STILL STUCK?
>
> Set Debug log detail to **4**, reproduce the issue and gather `fdn.log` with your game version, DLC and steps. See Warnings, logs & bug reports.

LI // FDN / LOGAIN INDUSTRIES

Credits, licence & links

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v1.00 BETA · X4: Foundations 9.00
FDN by **Logain Abler** (Logain Industries).

A new mod for X4, similar in idea to X3 FDN; not a port.

Thanks

SirNukes — SirNukes Mod Support APIs, the required dependency providing the Simple Menu API and Lua loader.

Licence

MIT License
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See `LICENSE` in the extension root or the GitHub repository for the full terms. Distributed free of charge under the MIT License, which permits commercial use.

Download and project

GitHub
https://github.com/mattryan72/FDN_X4

Release downloads · fdn_x4_v1.00.zip
FDN 1.00 BETA · tag v1.00 · GitHub pre-release
https://github.com/mattryan72/FDN_X4/releases

Required dependency · SirNukes Mod Support APIs
steamcommunity.com/sharedfiles/filedetails/?id=2042901274

Steam Workshop
(Workshop URL pending publication)

Egosoft forum · X4: Foundations – Scripts and Modding
(forum thread URL pending publication)

> UNOFFICIAL FAN CONTENT
>
> Requires a legally owned copy of X4: Foundations. No extracted game files are included. Screenshots are supplied gameplay captures; no extracted game assets are redistributed.

Guide documentation snapshot: public README.md and CHANGELOG.md at 6b1d9ea; checked shipped defaults supplied for v1.00 BETA. Pre-release plan: tag v1.00, title FDN 1.00 BETA; asset fdn_x4_v1.00.zip. Fonts: Orbitron, Rajdhani, Exo 2 and Share Tech Mono (SIL Open Font License). PDF intended for GitHub distribution.